

DIGITAL SAT MATH MASTERY SERIES

DATA INTERPRETATION FROM TABLES AND GRAPHS

Total Questions: 50 | Easy: 8 | Medium: 19 | Hard: 23

DOCUMENT 1 — PRACTICE QUESTIONS

Aligned to Digital SAT Question Style & Cognitive Demand

All questions original — fresh numbers, fresh contexts

Topic	Data Interpretation from Tables and Graphs
Document	DOCUMENT 1 — PRACTICE QUESTIONS
Questions	50
Difficulty	Easy: 8 Medium: 19 Hard: 23
Format	Multiple Choice (4 options)
Exam Target	Digital SAT (DSAT)

DATA INTERPRETATION FROM TABLES AND GRAPHS — PRACTICE QUESTIONS

Instructions: Choose the best answer for each question. All questions are original and written in authentic Digital SAT style. Difficulty level is noted for each question.

Q1. DIFFICULTY: **EASY**

The table below shows the number of visitors to a museum each day for one week. Day | Mon | Tue | Wed | Thu | Fri | Sat | Sun Visitors | 142 | 98 | 115 | 201 | 188 | 320 | 275 What was the total number of visitors on the two highest-attendance days?

- A) 595
- B) 508
- C) 623
- D) 520

Q2. DIFFICULTY: **EASY**

The bar graph below represents the number of books read by five students during summer. Student | Books Read Ana | 8 Ben | 5 Carlos | 12 Diana | 7 Ethan | 10 Which student read exactly twice as many books as Ben?

- A) Ana
- B) Carlos
- C) Diana
- D) Ethan

Q3. DIFFICULTY: **EASY**

The table shows the monthly rainfall (in inches) for a city over four months. Month | Jan | Feb | Mar | Apr Rainfall | 3.2 | 2.8 | 4.1 | 1.9 What is the average (mean) monthly rainfall over these four months?

- A) 3.0 inches
- B) 2.95 inches
- C) 3.25 inches
- D) 2.85 inches

Q4. DIFFICULTY: **EASY**

A scatterplot shows the relationship between hours studied (x) and test score (y). The line of best fit passes through the points (2, 65) and (8, 95). What is the slope of the line of best fit?

- A) 4
- B) 5
- C) 6
- D) 3

Q5. *DIFFICULTY: EASY*

The table below shows the price of gasoline at a station over five weeks. Week | 1 | 2 | 3 | 4 | 5 Price | 3.20 | 3.35 | 3.50 | 3.65 | 3.80 By how much did the price increase from Week 1 to Week 5?

- A) \$0.50
 - B) \$0.60
 - C) \$0.65
 - D) \$0.45
-

Q6. *DIFFICULTY: EASY*

A pie chart shows how a student spends 24 hours in a day: Sleep 37.5%, School 25%, Homework 12.5%, Recreation 16.7%, Other 8.3%. How many hours per day does the student spend sleeping?

- A) 8 hours
 - B) 9 hours
 - C) 10 hours
 - D) 7 hours
-

Q7. *DIFFICULTY: EASY*

The table below shows the population of four towns. Town | Alton | Bexley | Croft | Dune Population | 4,820 | 7,310 | 3,095 | 5,640 Which town has the second-largest population?

- A) Alton
 - B) Bexley
 - C) Croft
 - D) Dune
-

Q8. *DIFFICULTY: EASY*

A line graph shows a company's quarterly profits (in thousands of dollars): Q1: \$45k, Q2: \$52k, Q3: \$38k, Q4: \$61k. Between which two consecutive quarters did profit increase the most?

- A) Q1 to Q2
 - B) Q2 to Q3
 - C) Q3 to Q4
 - D) Q1 to Q4
-

Q9. *DIFFICULTY: MEDIUM*

The table below shows data on plant growth. Days of sunlight per week | 1 | 2 | 3 | 4 | 5 | 6 | 7 Height after 4 weeks (cm) | 4 | 7 | 10 | 14 | 18 | 21 | 25 A line of best fit is drawn. Based on the linear trend, what is the approximate height of a plant that receives 3.5 days of sunlight per week?

- A) 11 cm
 - B) 12 cm
 - C) 13 cm
 - D) 10 cm
-

Q10. DIFFICULTY: **MEDIUM**

The two-way table below shows survey results from 200 students about their preferred study location. | Library | Home | Café | Total Freshmen | 24 | 36 | 15 | 75 Sophomores | 18 | 42 | 20 | 80 Juniors | 12 | 22 | 11 | 45 Total | 54 | 100 | 46 | 200 What percentage of students who prefer the library are freshmen?

- A) 32%
 - B) 44%
 - C) 45%
 - D) 48%
-

Q11. DIFFICULTY: **MEDIUM**

The scatterplot below displays data for 10 cars, showing engine size (liters) on the x-axis and fuel efficiency (mpg) on the y-axis. The line of best fit has equation $y = -6x + 42$. According to this model, what is the predicted fuel efficiency for a car with a 4.5-liter engine?

- A) 12 mpg
 - B) 15 mpg
 - C) 18 mpg
 - D) 9 mpg
-

Q12. DIFFICULTY: **MEDIUM**

A table shows temperature ($^{\circ}\text{C}$) and the number of hot drinks sold at a café. Temperature ($^{\circ}\text{C}$) | 5 | 10 | 15 | 20 | 25 Hot drinks sold | 180 | 150 | 120 | 90 | 60 Which equation best models this relationship, where H is drinks sold and T is temperature?

- A) $H = -6T + 210$
 - B) $H = 6T + 150$
 - C) $H = -6T + 150$
 - D) $H = -30T + 210$
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Q13. DIFFICULTY: **MEDIUM**

The table below shows the scores of 8 students on a math quiz. Student | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 Score | 72 | 85 | 91 | 68 | 77 | 95 | 83 | 79 What is the median score?

- A) 82
 - B) 81
 - C) 80
 - D) 83
-

Q14. DIFFICULTY: **MEDIUM**

A researcher plotted income (x , in thousands of dollars) vs. years of education (y). The line of best fit is $y = 0.4x - 2$. What does the x -intercept of this line represent in this context?

- A) The income at which education level is zero
 - B) The education level when income is zero
 - C) The income level at which the model predicts 0 years of education
 - D) The slope of the relationship
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Q15. DIFFICULTY: **MEDIUM**

The table below tracks a runner's mile times over 6 weeks of training. Week | 1 | 2 | 3 | 4 | 5 | 6 Mile time | 9:12 | 9:00 | 8:48 | 8:36 | 8:24 | 8:12 (min:sec) If the linear trend continues, what would the runner's mile time be in Week 8?

- A) 7:48
 - B) 7:36
 - C) 8:00
 - D) 7:54
-

Q16. DIFFICULTY: **MEDIUM**

The following table shows the relationship between advertising spending (in thousands of dollars) and weekly sales revenue (in thousands of dollars) for a company. Ad Spend | 2 | 4 | 6 | 8 | 10 Revenue | 18 | 26 | 34 | 42 | 50 Which of the following best describes the relationship?

- A) For every \$1,000 increase in advertising, revenue increases by \$8,000
 - B) For every \$1,000 increase in advertising, revenue increases by \$4,000
 - C) Revenue is approximately 9 times the advertising spend
 - D) For every \$2,000 increase in advertising, revenue increases by \$4,000
-

Q17. DIFFICULTY: **MEDIUM**

The two-way table below shows whether students passed or failed a standardized test, broken down by whether they attended a test prep course. | Passed | Failed | Total Attended prep | 85 | 15 | 100 No prep | 42 | 58 | 100 Total | 127 | 73 | 200 What is the difference in pass rates between students who attended prep and those who did not?

- A) 43 percentage points
 - B) 43%
 - C) 44 percentage points
 - D) 45%
-

Q18. DIFFICULTY: **MEDIUM**

A graph shows the altitude of a plane (in feet) as a function of time (in minutes) since takeoff. The altitude increases linearly from 0 to 30,000 feet over 20 minutes, then remains constant for 80 minutes, then decreases linearly to 0 over 15 minutes. What is the rate of altitude increase during takeoff?

- A) 1,500 feet per minute
 - B) 2,000 feet per minute
 - C) 1,200 feet per minute
 - D) 1,800 feet per minute
-

Q19. DIFFICULTY: **MEDIUM**

The table shows monthly subscription counts for a streaming service over 6 months. Month | 1 | 2 | 3 | 4 | 5 | 6 Subscriptions | 320 | 384 | 461 | 553 | 664 | 797 (thousands) A student claims the data shows a linear relationship. Is this claim supported?

- A) Yes, because the values keep increasing
 - B) No, because the differences between consecutive months are not constant
 - C) Yes, because all values are positive
 - D) No, because the values exceed 700
-

Q20. DIFFICULTY: **MEDIUM**

The scatterplot below shows the number of hours of TV watched per day (x) and GPA (y) for 15 students. The line of best fit is $y = -0.25x + 3.8$. A student watches 4 hours of TV per day. What GPA does the model predict?

- A) 2.8
 - B) 3.0
 - C) 2.4
 - D) 3.2
-

Q21. DIFFICULTY: **MEDIUM**

The bar graph below shows the number of medals won by five countries in an international competition. Country | Gold | Silver | Bronze Arovia | 8 | 5 | 7 Bental | 4 | 9 | 6 Corum | 11 | 3 | 8 Drex | 6 | 7 | 12 Efton | 3 | 11 | 4 Which country won the greatest total number of medals?

- A) Arovia
 - B) Bental
 - C) Corum
 - D) Drex
-

Q22. DIFFICULTY: **MEDIUM**

A line graph shows a city's average temperature ($^{\circ}\text{F}$) each month. The highest point is 88°F in July and the lowest is 24°F in January. What is the range of temperatures shown?

- A) 64°F
 - B) 62°F
 - C) 66°F
 - D) 60°F
-

Q23. DIFFICULTY: **HARD**

The table below shows data from a study on hours of sleep and reaction time. Hours of sleep | 4 | 5 | 6 | 7 | 8 | 9 Reaction time | 420 | 380 | 340 | 300 | 260 | 220 (milliseconds) According to the linear model, what would the predicted reaction time be for a person who gets 10 hours of sleep?

- A) 180 ms
 - B) 200 ms
 - C) 160 ms
 - D) 220 ms
-

Q24. DIFFICULTY: **HARD**

The two-way table below shows the results of a survey of 500 adults about exercise habits and reported health status. | Good health | Fair health | Poor health | Total Exercises daily | 145 | 35 | 20 | 200 Exercises weekly | 110 | 55 | 35 | 200 Rarely exercises | 40 | 55 | 5 | 100 Total | 295 | 145 | 60 | 500 What fraction of those in good health exercises daily?

- A) $145/295$
 - B) $145/200$
 - C) $29/59$
 - D) $145/500$
-

Q25. DIFFICULTY: **HARD**

The scatterplot below shows data comparing outdoor temperature (x , in $^{\circ}\text{F}$) and the number of coats sold at a store (y). The line of best fit passes through (20, 85) and (60, 25). Which of the following is closest to the number of coats predicted to be sold when the temperature is 45°F ?

- A) 47 coats
 - B) 48 coats
 - C) 50 coats
 - D) 52 coats
-

Q26. DIFFICULTY: **HARD**

The table shows the value of a machine over time. Year | 0 | 2 | 4 | 6 | 8 Value(\$) | 50,000 | 40,000 | 30,000 | 20,000 | 10,000 According to the linear model, in which year will the machine's value first reach \$0?

- A) Year 10
- B) Year 11
- C) Year 12
- D) Year 9

Q27. DIFFICULTY: **HARD**

A two-way table shows enrollment in elective courses at a school. | Art | Music | Drama | Total Grade 10 | 48 | 32 | 20 | 100 Grade 11 | 36 | 44 | 30 | 110 Grade 12 | 24 | 36 | 40 | 100 Total | 108 | 112 | 90 | 310 If a student is randomly selected from those enrolled in Drama, what is the probability that the student is in Grade 12?

- A) 4/9
- B) 40/90
- C) 4/10
- D) 40/310

Q28. DIFFICULTY: **HARD**

The line graph below shows electricity usage (in kilowatt-hours) for a household each month. The values are: Jan: 920, Feb: 880, Mar: 740, Apr: 620, May: 510, Jun: 480, Jul: 530, Aug: 490, Sep: 560, Oct: 670, Nov: 810, Dec: 950. In which month-to-month interval did usage decrease the most?

- A) February to March
- B) March to April
- C) April to May
- D) January to February

Q29. DIFFICULTY: **HARD**

The table below shows the results of a clinical trial. | Recovered | Did not recover | Total Treatment | 132 | 68 | 200 Placebo | 84 | 116 | 200 Total | 216 | 184 | 400 Which of the following statements is best supported by the table?

- A) The treatment was not effective since most patients in both groups recovered
 - B) The treatment group had a recovery rate 24 percentage points higher than the placebo group
 - C) The placebo was equally effective as the treatment
 - D) The treatment doubled the recovery rate compared to placebo
-

Q30. DIFFICULTY: **HARD**

The scatterplot below shows test scores before and after a tutoring program for 12 students. The line $y = x$ is also shown. A point significantly above the line $y = x$ indicates what?

- A) The student's after score is higher than before
 - B) The student's before score is higher than after
 - C) The student scored the same before and after
 - D) The student did not complete the program
-

Q31. DIFFICULTY: **HARD**

The table shows data on population density (people per sq mile) and average commute time (minutes) for six cities. City | Density | Commute A | 2,400 | 28 B | 5,800 | 38 C | 1,200 | 22 D | 8,400 | 46 E | 3,600 | 32 F | 6,900 | 42 A researcher fits a linear model to this data. Approximately how many people per square mile would correspond to a predicted commute time of 35 minutes, based on the trend in the data?

- A) About 4,500
 - B) About 5,000
 - C) About 4,800
 - D) About 5,200
-

Q32. DIFFICULTY: **HARD**

The table below shows results from a consumer survey on two smartphone brands. | Satisfied | Dissatisfied | Total Brand X users | 180 | 60 | 240 Brand Y users | 120 | 40 | 160 Total | 300 | 100 | 400 A reporter claims Brand X has a higher satisfaction rate than Brand Y. Is this claim correct?

- A) Yes, because Brand X has more satisfied customers
 - B) No, because both brands have a 75% satisfaction rate
 - C) Yes, because Brand X has 240 customers
 - D) No, because Brand Y has more satisfied customers per user
-

Q33. DIFFICULTY: **HARD**

A histogram shows the distribution of ages of participants in a marathon. The bins are 20-29, 30-39, 40-49, 50-59, 60-69 with frequencies 45, 112, 98, 74, 21, respectively. Approximately what percentage of participants are between 30 and 49 years old?

- A) About 60%
 - B) About 55%
 - C) About 65%
 - D) About 48%
-

Q34. DIFFICULTY: **HARD**

The line of best fit for a data set is $y = 2.5x + 4$. One data point is (6, 22). What is the residual for this data point?

- A) 3
- B) 4
- C) 2
- D) -3

Q35. DIFFICULTY: **HARD**

The table below shows production output (units per hour) and defect rate (%) for a factory over 8 shifts. Output | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 110 Defect | 2 | 3 | 3 | 4 | 5 | 6 | 8 | 10 According to the trend, if output increases to 120 units per hour, what is the predicted defect rate?

- A) 11%
- B) 12%
- C) 13%
- D) 10%

Q36. DIFFICULTY: **HARD**

The two-way table below records results of a study on coffee consumption and sleep quality for 300 participants. | Good sleep | Poor sleep | Total High caffeine | 30 | 120 | 150 Low caffeine | 120 | 30 | 150 Total | 150 | 150 | 300 Which of the following best describes the association?

- A) There is no association between caffeine and sleep quality
- B) High caffeine intake is associated with poor sleep quality
- C) Low caffeine intake is associated with poor sleep quality
- D) Caffeine intake does not affect sleep quality based on this sample

Q37. DIFFICULTY: **HARD**

The scatterplot shows sales (y, thousands) vs. price (x, dollars) with a line of best fit $y = -0.8x + 120$. A store currently sells at \$90. What is the predicted difference in sales if the price is reduced to \$70?

- A) 16 thousand units
 - B) 12 thousand units
 - C) 8 thousand units
 - D) 20 thousand units
-

Q38. DIFFICULTY: **MEDIUM**

The table below shows ticket sales for a concert over 4 days before the event.

Days before event	1	2	3	4
Tickets sold	120	180	270	405

A student fits a linear model to this data. Another student notices the data might be exponential. Which student is more likely correct, and why?

- A) The linear student, because ticket sales increase each day
 - B) The exponential student, because each day's sales are approximately 1.5 times the previous day's
 - C) The linear student, because the differences are increasing
 - D) The exponential student, because ticket sales will eventually stop
-

Q39. DIFFICULTY: **MEDIUM**

The graph below shows the speed of a car (in mph) over a 5-hour trip. The car travels at 60 mph for the first 2 hours, then slows to 30 mph for 1 hour, then stops for 30 minutes, then resumes at 50 mph for 1.5 hours. What is the total distance traveled?

- A) 225 miles
 - B) 225.5 miles
 - C) 200 miles
 - D) 210 miles
-

Q40. DIFFICULTY: **MEDIUM**

The table below shows survey data on preferred social media platform by age group.

	App A	App B	App C	Total
Ages 13-17	90	40	20	150
Ages 18-24	60	80	60	200
Ages 25-34	30	70	100	200
Ages 35+	20	50	80	150
Total	200	240	260	700

What proportion of users aged 13-17 prefer App A?

- A) $\frac{3}{5}$
 - B) $\frac{90}{200}$
 - C) $\frac{9}{70}$
 - D) $\frac{90}{700}$
-

Q41. DIFFICULTY: **MEDIUM**

A line graph shows a student's reading speed (words per minute) at the beginning of each school month: Sep: 140, Oct: 155, Nov: 168, Dec: 180, Jan: 195, Feb: 209. Which of the following is closest to the average monthly increase in reading speed?

- A) About 13.8 wpm
 - B) About 14 wpm
 - C) About 15 wpm
 - D) About 12 wpm
-

Q42. DIFFICULTY: **MEDIUM**

The table below shows annual household income (in thousands of dollars) and annual savings (in thousands of dollars) for 6 households. Income | 40 | 55 | 70 | 85 | 100 | 120 Savings | 2 | 5 | 8 | 12 | 17 | 24 A student claims savings increase linearly with income. Is this supported?

- A) Yes, because savings always increase
 - B) No, because the increases in savings are not constant as income rises by equal amounts
 - C) Yes, because the relationship is positive
 - D) No, because some incomes are more than triple others
-

Q43. DIFFICULTY: **HARD**

The dot plot below represents the number of goals scored per game by a soccer team over a 20-game season. Goals | 0 | 1 | 2 | 3 | 4 | 5 Games | 2 | 5 | 7 | 4 | 1 | 1 What is the mean number of goals per game for this season?

- A) 2.0
 - B) 2.1
 - C) 1.9
 - D) 2.2
-

Q44. DIFFICULTY: **HARD**

A two-way table from a study on diet and cholesterol for 400 adults: | High cholesterol | Normal cholesterol | Total High-fat diet | 120 | 80 | 200 Low-fat diet | 60 | 140 | 200 Total | 180 | 220 | 400 What is the relative risk of high cholesterol for high-fat diet compared to low-fat diet?

- A) 2.0
 - B) 1.5
 - C) 2.5
 - D) 1.8
-

Q45. DIFFICULTY: **HARD**

The table below shows average monthly temperatures (°F) in two cities. Month | Jan | Feb | Mar | Apr | May | Jun City A | 28 | 32 | 42 | 54 | 65 | 74 City B | 52 | 55 | 60 | 66 | 72 | 78 In which month is the difference in temperatures between the two cities the greatest?

- A) January
 - B) February
 - C) March
 - D) April
-

Q46. DIFFICULTY: **HARD**

The scatterplot shows years of experience (x) and salary (y , thousands) with line of best fit $y = 3.2x + 38$. A data point at (8, 65) has a residual of:

- A) 1.4
 - B) -1.4
 - C) 0.6
 - D) -0.6
-

Q47. DIFFICULTY: **HARD**

The table below shows the number of defective items produced per 1,000 at a factory, before and after a quality improvement program.

Month	Jan	Feb	Mar	Apr	May	Jun
Before	42	39	45	38	41	40
After	28	25	30	22	27	24

What is the mean reduction in defective items per 1,000 after the program?

- A) 15
 - B) 14.5
 - C) 15.5
 - D) 16
-

Q48. DIFFICULTY: **HARD**

The bar graph shows the percentage of students scoring in each range on a standardized test.

Range	Below 400	400-499	500-599	600-699	700-800
Percent	8%	18%	34%	27%	13%

If 2,500 students took the test, approximately how many scored 600 or above?

- A) 900
 - B) 1,000
 - C) 950
 - D) 850
-

Q49. DIFFICULTY: **HARD**

The table shows high and low temperatures ($^{\circ}\text{F}$) for a city for 5 days.

Day	High	Low
Mon	78	55
Tue	82	58
Wed	75	52
Thu	89	63
Fri	71	50

What was the mean daily temperature range (high minus low) across all 5 days?

- A) 23.4°F
 - B) 24°F
 - C) 23°F
 - D) 25°F
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Q50. DIFFICULTY: **HARD**

The line of best fit for a data set showing monthly revenue (y , in thousands) as a function of months since launch (x) is $y = 4.8x + 12$. A company uses this model to predict revenue in month 18. However, the actual revenue in month 18 is \$91,000. Which of the following is true?

- A) The actual revenue exceeds the predicted revenue by \$7,400
 - B) The predicted revenue exceeds the actual revenue by \$7,400
 - C) The actual revenue exceeds the predicted revenue by \$7,000
 - D) The model is accurate for month 18
-